Cross-Dimensional Equivalence of Offset Clustering: A Framework for Teacher–Student Comparison

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1 Setup

Throughout, $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm and $\langle\cdot,\cdot\rangle$ the standard Euclidean inner product. We work with two parameter tuples

$$\theta_T = (E_T, W^T_{\text{gate}}, W^T_{\text{down}}) \in \mathbb{R}^{|V|\times d_T} \times \mathbb{R}^{d^T_{\text{ffn}}\times d_T} \times \mathbb{R}^{d^T_{\text{ffn}}\times d_T},$$

$$\theta_S = (E_S, W^S_{\text{gate}}, W^S_{\text{down}}) \in \mathbb{R}^{|V|\times d_S} \times \mathbb{R}^{d^S_{\text{ffn}}\times d_S} \times \mathbb{R}^{d^S_{\text{ffn}}\times d_S},$$

representing two transformer language models that share a tokenizer (and hence vocabulary V) but may differ in residual-stream dimension $d_S \le d_T$ and FFN feature count $d^S_{\text{ffn}}, d^T_{\text{ffn}}$. We refer to θ_T as the *teacher* and θ_S as the *student*, motivated by the distillation setting where the student is intended to be a compressed representation of the teacher.

We adopt the anchoring, offset, and pair-partition definitions of the same-dimensional setting (Section 2 of the companion note), specialized to each parameter tuple. In particular:

- $\sigma_\theta(f) = \arg\max_{t\in V^{\text{word}}} \operatorname{cossim}(E[t], g^\theta_f)$ is the source anchor.
- $\tau_\theta(f) = \arg\max_{t\in V} \langle E[t], w^\theta_f\rangle$ is the target anchor.
- $\mathcal{O}(f;\theta) = \widehat{E[\tau_\theta(f)] - E[\sigma_\theta(f)]} \in S^{d-1}$ is the unit offset.
- $\mu^\theta_f = \|E[\tau_\theta(f)] - E[\sigma_\theta(f)]\|$ is the unnormalized offset magnitude.
- $\mathcal{P}_\theta$ is the pair partition: $f \sim_P f'$ iff $(\sigma_\theta(f), \tau_\theta(f)) = (\sigma_\theta(f'), \tau_\theta(f'))$.
- $n_{[p]}$ is the multiplicity of pair class $[p]$.

2 Cross-dimensional projection

Definition 2.1 (Cross-dimensional projection)**.** *A* cross-dimensional projection *from $\mathbb{R}^{d_T}$ to $\mathbb{R}^{d_S}$ is a matrix $P \in \mathbb{R}^{d_S\times d_T}$ satisfying $PP^\top = I_{d_S}$. Equivalently, the rows of P form an orthonormal set in $\mathbb{R}^{d_T}$, or $P^\top$ is a Stiefel matrix in $\mathbb{R}^{d_T\times d_S}$.*

Definition 2.2 (Null space and orthogonal decomposition)**.** *The* null space *of P is $N(P) = \{v \in \mathbb{R}^{d_T} : Pv = 0\}$, with $\dim N(P) = d_T - d_S$. Every $v \in \mathbb{R}^{d_T}$ admits the unique orthogonal decomposition*

$$v = v_\| + v_\perp, \qquad v_\| \in N(P), \quad v_\perp \in N(P)^\perp,$$

where the orthogonality is with respect to the standard Euclidean inner product on $\mathbb{R}^{d_T}$. We refer to $v_\|$ as the null-space component *and $v_\perp$ as the* retained component*.*

Remark 2.3 (Relation to same-dimensional case)**.** *When $d_T = d_S$, P is square and $PP^\top = I_{d_S}$ implies $P \in O(d)$. The null space is trivial and every vector is its own retained component. The cross-dimensional framework reduces to the orthogonal-equivalence framework of the companion note as a special case.*

Remark 2.4 (Terminology note)**.** *We use "null space" rather than "kernel" throughout to avoid ambiguity with the kernel-methods sense of the term used in representation-similarity metrics such as CKA (**?**).*

Definition 2.5 (Cross-dimensional equivalence)**.** *Parameter tuples θ_T and θ_S are* cross-dimensionally equivalent *via a cross-dimensional projection P if*

$$E_T P^\top = E_S, \qquad W^T_{\text{gate}} P^\top = W^S_{\text{gate}}, \qquad W^T_{\text{down}} P^\top = W^S_{\text{down}},$$

with the right-multiplication convention that each row of E_T, W^T_{gate}, W^T_{down} is independently transformed.

Remark 2.6 (Restrictiveness of exact cross-dimensional equivalence)**.** *Definition **??** requires every embedding row, gate vector, and down-projection vector of the teacher to lie in $N(P)^\perp$ exactly. This is restrictive: it forces the entire teacher representation to be confined to a d_S-dimensional subspace of $\mathbb{R}^{d_T}$. In practice no nontrivial teacher–student pair satisfies this exactly; the framework is set up this way so that exact statements can be proven and then weakened in two stages: (i) by working at the multiset level rather than pointwise (Section **??**), and (ii) by quantifying the per-feature failure modes of the exactness hypothesis (Section **??**).*

3 The high-multiplicity multiset

The exact pointwise cross-dimensional equivariance of anchoring, offsets, and clustering — analogous to Theorem 3.3 of the same-dimensional case — requires the strong hypothesis of Definition **??**. We instead develop a coarser comparison object that does not require pointwise feature equivariance: the multiset of high-multiplicity pair classes.

Definition 3.1 (High-multiplicity pair-class multiset)**.** *For a parameter tuple θ, the* high-multiplicity pair-class multiset *is*

$$\mathcal{H}_\theta = \big\{([p], n_{[p]}) : [p] \in \mathcal{P}_\theta,\ n_{[p]} \geq 2\big\},$$

viewed as a finite multiset of (token-pair, multiplicity) records on $(V \times V) \times \mathbb{N}_{\geq 2}$.

Remark 3.2 (Why high multiplicity)**.** *Pair classes with multiplicity $n_{[p]} = 1$ correspond to single FFN features anchored on a unique source–target pair. These are idiosyncratic and highly sensitive to feature-level reorganization. Pair classes with $n_{[p]} \geq 2$ represent commitments by multiple FFN features to the same anchored relation, and are the structurally meaningful objects from the perspective of the content-coherence diagnostic of (**?**): a cluster contains a high-multiplicity pair class if and only if it contains repeated source–target pairs. The multiset $\mathcal{H}_\theta$ is the natural comparison object across teacher–student pairs because (i) it is invariant under feature-index relabeling, (ii) it is invariant under any deterministic clustering rule (by Proposition 2.2 of the companion note), and (iii) it directly captures the content-coherence signal.*

Definition 3.3 (High-multiplicity discrepancy)**.** *For parameter tuples θ_T and θ_S, write $n^T_{[p]}$ and $n^S_{[p]}$ for the multiplicities of the same token-pair $[p] \in V \times V$ in $\mathcal{P}_{\theta_T}$ and $\mathcal{P}_{\theta_S}$ respectively (with $n_{[p]} = 0$ if no feature anchors on $[p]$). The* high-multiplicity discrepancy *is*

$$\Delta_{\geq 2}(\theta_T, \theta_S) = \big|\{[p] : n^T_{[p]} \geq 2,\, n^S_{[p]} < 2\}\big| \ + \ \big|\{[p] : n^S_{[p]} \geq 2,\, n^T_{[p]} < 2\}\big|.$$

This counts pair classes that are high-multiplicity in exactly one of θ_T, θ_S.

4 At-risk features and the weak-form theorem

We now identify the per-feature failure mode of the cross-dimensional projection. The argument follows the structure of Lemma 4.4 in the companion note (anchoring stability under additive perturbation), with the perturbation magnitudes replaced by null-space components.

Lemma 4.1 (Cross-dimensional anchoring stability)**.** *Let θ_T and θ_S be cross-dimensionally equivalent via P in the sense of Definition **??**, except that the embedding rows and gate/down vectors are not assumed to*

lie in $N(P)^{\perp}$. Fix $f \in \mathcal{F}_T$, and let $\gamma_f^{\sigma}, \gamma_f^{\tau}$ denote the source and target anchoring margins computed in θ_T (Definition 4.6 of the companion note). Define

$$\nu_f^{\sigma} \;=\; \frac{\|E_T[\sigma_{\theta_T}(f)]_{\|}\|}{\|E_T[\sigma_{\theta_T}(f)]\|} \;+\; \frac{\|g_{f,\|}^{\theta_T}\|}{\|g_f^{\theta_T}\|} \;+\; R_f^{\sigma},$$

$$\nu_f^{\tau} \;=\; M_E \cdot \|w_{f,\|}^{\theta_T}\| \;+\; \|w_f^{\theta_T}\| \cdot \max_{t \in V} \|E_T[t]_{\|}\| \;+\; R_f^{\tau},$$

where $M_E = \max_{t \in V} \|E_T[t]\|$ and R_f^{σ}, R_f^{τ} collect the second-order contributions of orders $(\|E_T[\cdot]_{\|}\|/\|E_T[\cdot]\|)^2$ and $(\|g_{f,\|}^{\theta_T}\|/\|g_f^{\theta_T}\|)^2$ respectively. If

$$\nu_f^{\sigma} \;<\; \frac{\gamma_f^{\sigma}}{2} \qquad \text{and} \qquad \nu_f^{\tau} \;<\; \frac{\gamma_f^{\tau}}{2},$$

then $\sigma_{\theta_S}(f) = \sigma_{\theta_T}(f)$ and $\tau_{\theta_S}(f) = \tau_{\theta_T}(f)$.

To be supplied. The proof structure mirrors Lemma 4.4 of the companion note: bound the cosine and inner-product changes

□

Definition 4.2 (At-risk feature set)**.** *A teacher feature f is* at-risk *under cross-dimensional projection P if*

$$\nu_f^{\sigma} \geq \frac{\gamma_f^{\sigma}}{2} \quad \text{or} \quad \nu_f^{\tau} \geq \frac{\gamma_f^{\tau}}{2}.$$

The at-risk set *is*

$$\mathcal{F}_{\text{risk}}(P) = \{f \in \mathcal{F}_T : f \text{ is at-risk under } P\}.$$

*By Lemma **??**, every $f \in \mathcal{F}_T \setminus \mathcal{F}_{\text{risk}}(P)$ has its source and target anchoring preserved when passing to θ_S.*

Lemma 4.3 (Offset preservation off the at-risk set)**.** *Let θ_T and θ_S be cross-dimensionally equivalent via P. For every $f \in \mathcal{F}_T \setminus \mathcal{F}_{\text{risk}}(P)$, the unnormalized teacher offset $\widetilde{\mathcal{O}}(f;\theta_T) = E_T[\tau_{\theta_T}(f)] - E_T[\sigma_{\theta_T}(f)]$ satisfies*

$$\mathcal{O}(f;\theta_S) \;=\; \frac{\widetilde{\mathcal{O}}(f;\theta_T)P^{\top}}{\left\|\widetilde{\mathcal{O}}(f;\theta_T)P^{\top}\right\|}.$$

Furthermore, if $\widetilde{\mathcal{O}}(f;\theta_T) \in N(P)^{\perp}$, then $\left\|\widetilde{\mathcal{O}}(f;\theta_T)P^{\top}\right\| = \mu_f^{\theta_T}$ and the offset magnitude is preserved exactly.

*To be supplied. The first claim follows from Lemma **??** applied to anchor preservation, combined with the linearity of $P^{\top}$.*

□

Definition 4.4 (At-risk concentration in a pair class)**.** *For a pair class $[p] \in \mathcal{P}_{\theta_T}$ and cross-dimensional projection P, the* at-risk concentration *is*

$$\rho_{\text{risk}}([p];P) \;=\; \big|[p] \cap \mathcal{F}_{\text{risk}}(P)\big|.$$

The pair class $[p]$ is survivable *under P if $n_{[p]} - \rho_{\text{risk}}([p];P) \geq 2$, i.e., at least two of its features have anchoring preserved under projection.*

Theorem 4.5 (Cross-dimensional weak-form invariance)**.** *Let θ_T and θ_S be cross-dimensionally equivalent via cross-dimensional projection P (Definition **??**). Then the high-multiplicity discrepancy is bounded by*

$$\Delta_{\geq 2}(\theta_T, \theta_S) \;\leq\; \mathcal{T}(P) + \mathcal{S}(P),$$

where

$$\mathcal{T}(P) = \big|\{[p] \in \mathcal{P}_{\theta_T} : n_{[p]} \geq 2,\; n_{[p]} - \rho_{\text{risk}}([p];P) < 2\}\big|,$$
$$\mathcal{S}(P) = \big|\{[p'] \in \mathcal{P}_{\theta_S} : n_{[p']}^{S} \geq 2,\; [p'] \notin \mathcal{P}_{\theta_T}\}\big|.$$

The first term ($\mathcal{T}$) bounds high-multiplicity teacher pair classes that may fail to survive projection due to too many of their features being at-risk. The second term ($\mathcal{S}$) accounts for high-multiplicity pair classes created *in the student through reshuffling of at-risk teacher features.*

To be supplied. Sketch: Lemma **??** *guarantees that features outside* $\mathcal{F}_{\text{risk}}(P)$ *keep their teacher anchoring in* θ_S*. A teacher p with*$n_{[p]} \geq 2$ remains high-multiplicity in θ_S if at least 2 of its features are not at-risk, i.e., $n_{[p]} - \rho_{\text{risk}}([p]; P) \geq 2$. Pair classes failing this are the candidates for being lost; they are bounded by $\mathcal{T}(P)$. The symmetric contribution $\mathcal{S}(P)$ handles new high-multiplicity classes appearing in the student, which can only arise from reshuffled at-risk features, bounded by the count of student high-multiplicity classes not present in the teacher.] □

Remark 4.6 (Distributional sensitivity)**.** *The bound of Theorem* **??** *depends on how the at-risk features distribute across pair classes, not merely on the total* $|\mathcal{F}_{\text{risk}}(P)|$*. If at-risk features cluster within a small number of pair classes, those classes may collapse below the high-multiplicity threshold, contributing to* $\mathcal{T}(P)$*. If at-risk features spread across many pair classes, each class loses a small fraction and most remain high-multiplicity. This is structurally analogous to the distinction between global and per-cluster flip fractions in the perturbative analysis of the companion note (Remark 4.12), and motivates a corresponding empirical check: the distribution of* $\rho_{\text{risk}}([p]; P)/n_{[p]}$ *across high-multiplicity pair classes for a candidate* P*.*

Remark 4.7 (Empirical applicability)**.** *For a real teacher–student pair* (θ_T, θ_S)*, no exact cross-dimensional projection* P *relating them is given a priori. The empirical use of Theorem* **??** *requires first estimating the optimal* P *— e.g., via a Procrustes-style fit minimizing* $\|\theta_T P^\top - \theta_S\|_F$ *over Stiefel matrices* $P^\top$ *— and then evaluating the bound's right-hand side under that* P*. The framework treats the exact-equivalence assumption of Definition* **??** *as a base case; a perturbative version that accommodates a residual* $\theta_T P^\top \neq \theta_S$ *remains to be developed, paralleling the structure of Section 4 of the companion note.*

5 Open directions

- *Perturbative cross-dimensional invariance.* The current framework assumes exact cross-dimensional equivalence. A natural extension introduces a residual term $\Delta = \theta_S - \theta_T P^\top$ and quantifies the discrepancy bound in terms of both $\mathcal{F}_{\text{risk}}(P)$ and $\|\Delta\|$. The interaction between null-space load and additive perturbation should mirror Section 4 of the companion note.

- *Optimal cross-dimensional projection.* For a given teacher–student pair, the choice of P minimizing $\Delta_{\geq 2}$ is not necessarily the Frobenius-Procrustes optimum, since the latter does not account for anchoring margins. A principled criterion for selecting P in the empirical setting is open.

- *Distributional structure of* $\mathcal{F}_{\text{risk}}(P)$*.* Whether at-risk features concentrate in particular pair classes, or distribute uniformly, is empirically open and determines the tightness of Theorem **??**.

- *Connection to representation-manifold smoothness.* The linear cross-dimensional framework here is purely algebraic. Recent work treating LLM representations as smooth Riemannian submanifolds (**?**) suggests an alternative formulation in which P is replaced by a Riemannian submersion between latent semantic manifolds, with information loss quantified by the metric vertical component. This is a natural future direction with substantially heavier mathematical apparatus.

References

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